

CSA4001-MANAGEMENT INFORMATION SYSTEM FOR DATA OPTIMIZATION

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1. Draw a UML diagram for hotel reservation system. In a hotel reservation system, a customer can make online booking for a hotel, by specifying the accommodation requirements such as type of room (AC/Non-AC, One bed/two bed), total no of rooms, duration of stay. The system selects a suitable hotel as per customer's requirements. If a hotel is found then the availability of rooms in that hotel is checked. The charges are calculated for the selected requirement and these are acknowledged to the customer. If the customer is satisfactory about the selection made by the system, then he confirms the reservation.

AIM

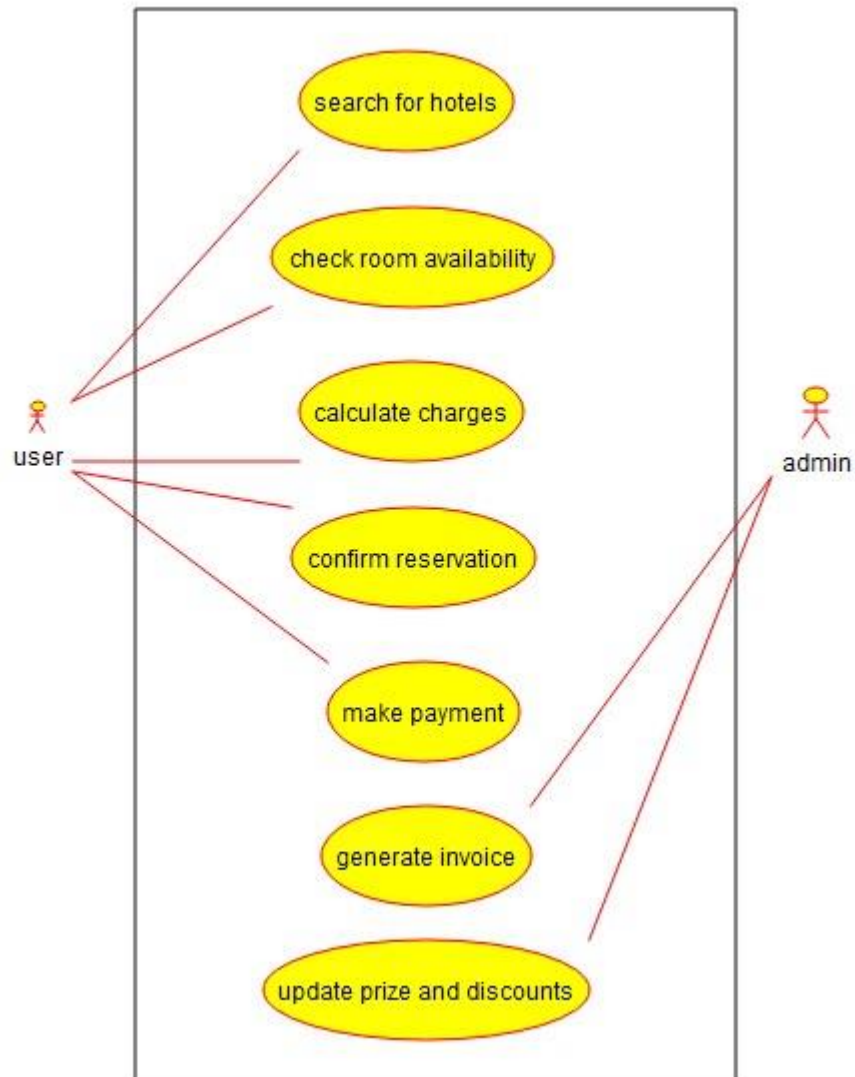
The aim is to design a **UML Activity Diagram** for a **Hotel Reservation System** to visualize the step-by-step process of booking a hotel room, checking availability, calculating charges, and confirming reservations.

PROCEDURE:

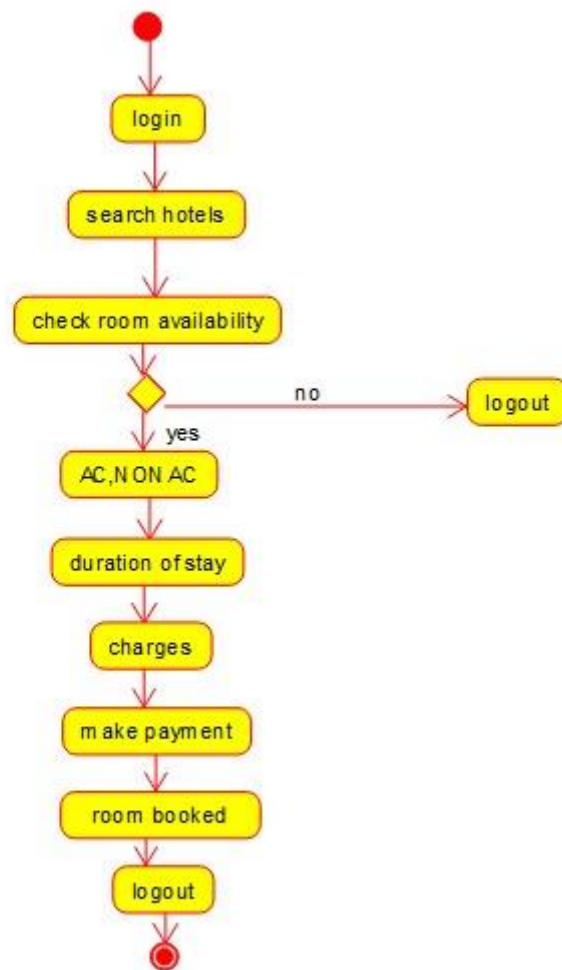
1. Identify the main activities: **Enter Booking Details, Search for Hotels, Check Room Availability, Calculate Charges, Show Charges to Customer, Confirm Reservation.**
2. Define the flow between activities, ensuring logical sequencing from customer input to reservation confirmation.
3. Identify decision points, such as **checking room availability** and **customer confirmation** before proceeding.
4. Establish actors involved in each activity, including **Customer** (provides details, confirms booking), **System** (searches hotels, calculates charges), and **Hotel Management** (updates room availability).
5. Use swimlanes to categorize activities under **Customer, System, and Hotel Management** for clarity.
6. Ensure branching and synchronization, such as an alternate flow if no suitable hotel is found.
7. Finalize the diagram, verifying completeness and correctness of the reservation process.

OUTPUT:

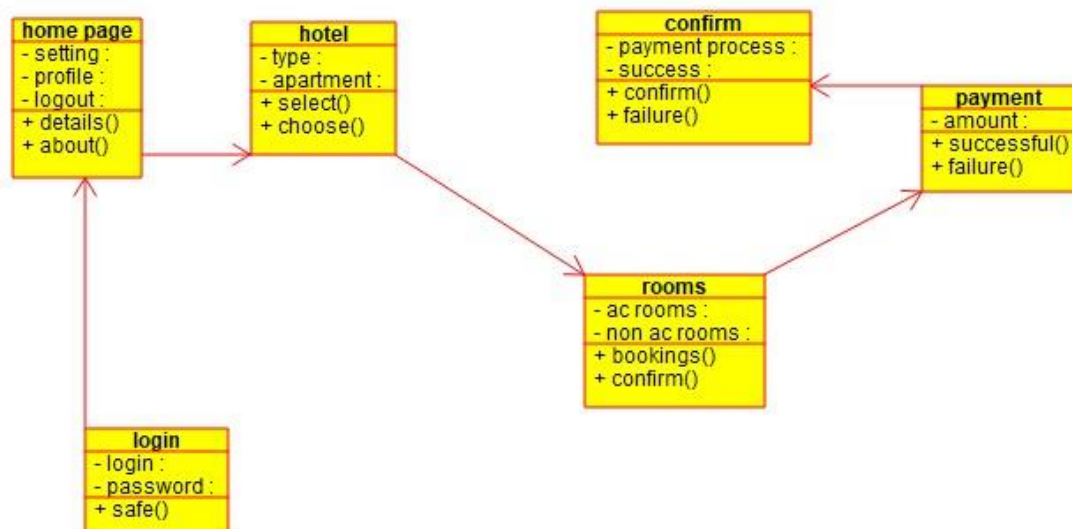
USE CASE DIAGRAM:



ACTIVITY DIAGRAM:



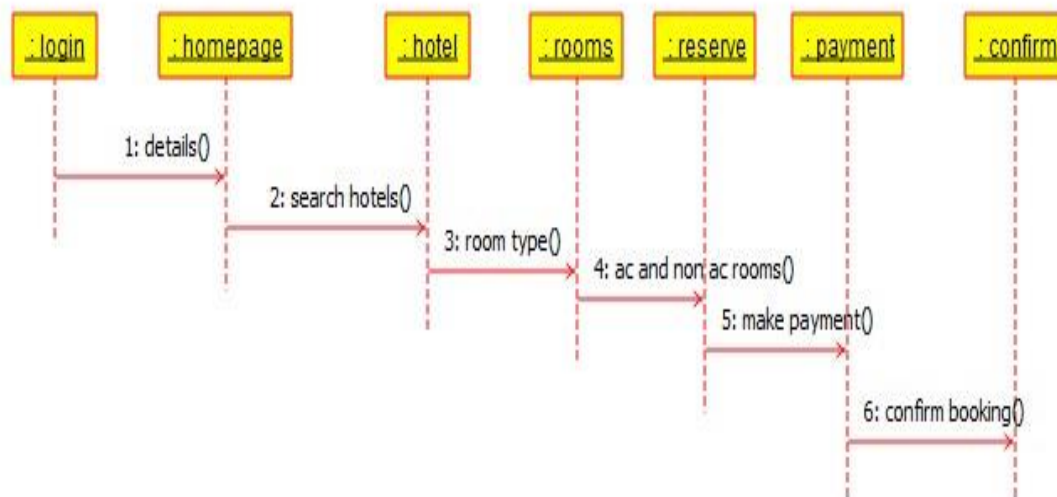
CLASS DIAGRAM:



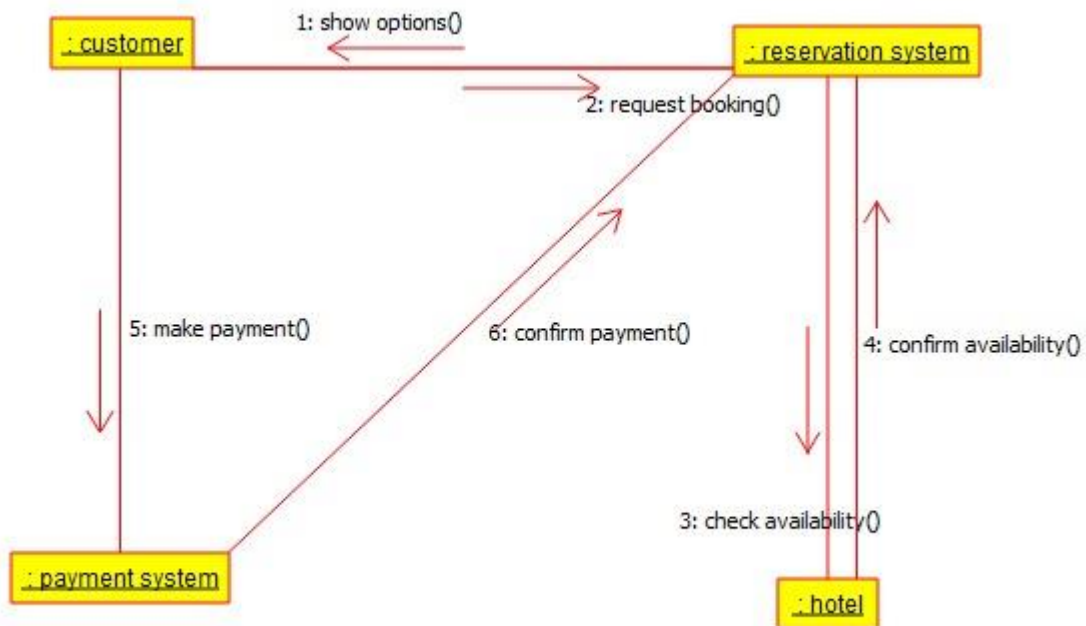
STATE DIAGRAM:



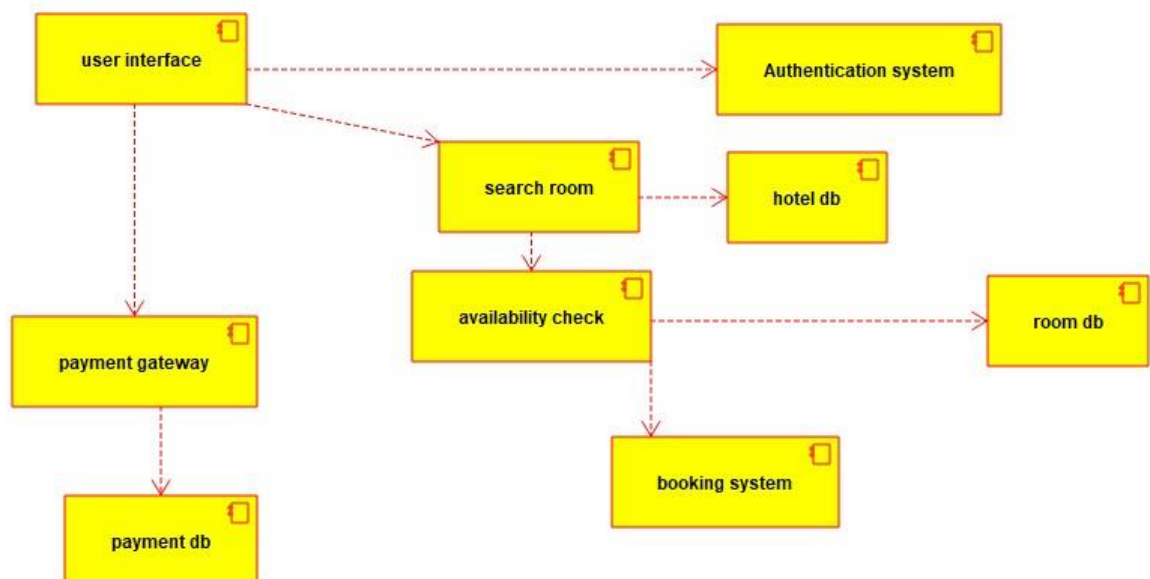
SEQUENCE DIAGRAM:



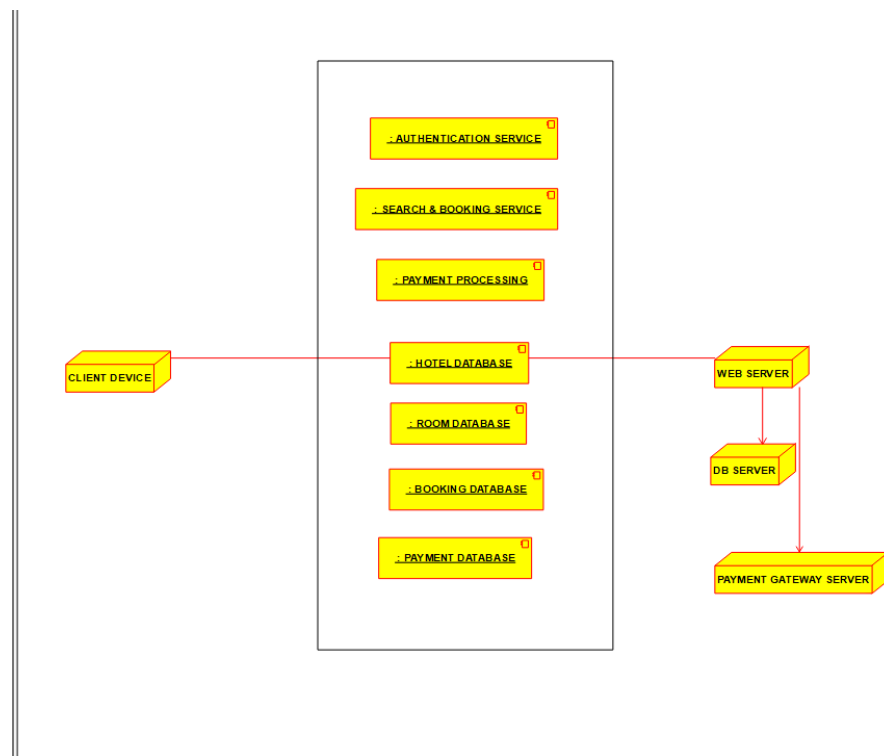
COMMUNICATION DIAGRAM:



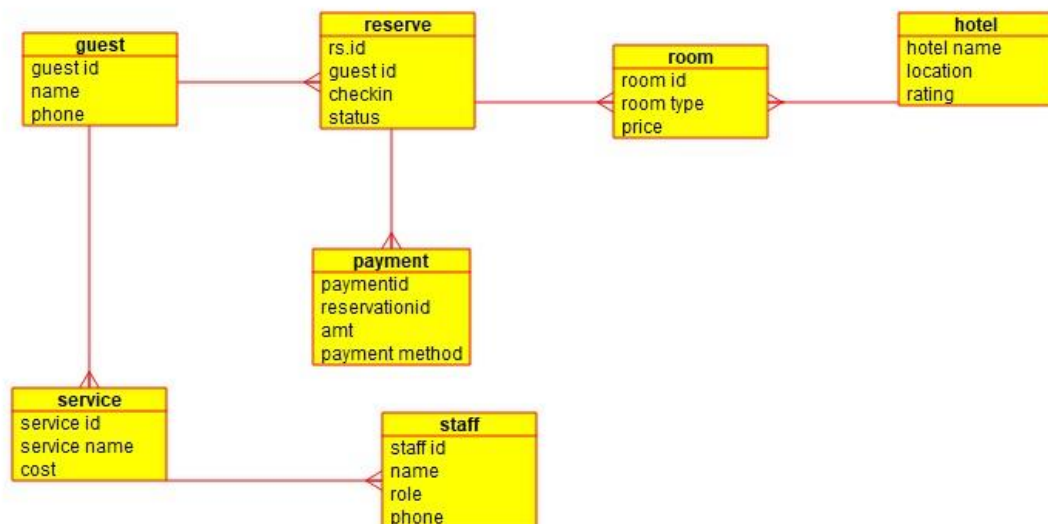
COMPONENT DIAGRAM:



DEPLOYMENT DIAGRAM:



ENTITY RELATIONSHIP DIAGRAM:



RESULT:

The Hotel Reservation System UML diagrams were successfully designed, covering all necessary interactions, workflows, and system components. This provides a structured representation of the system for implementation.